

Figure 8.2 Standard Dump Truck



Figure 8.3 Semitrailer Dump Truck

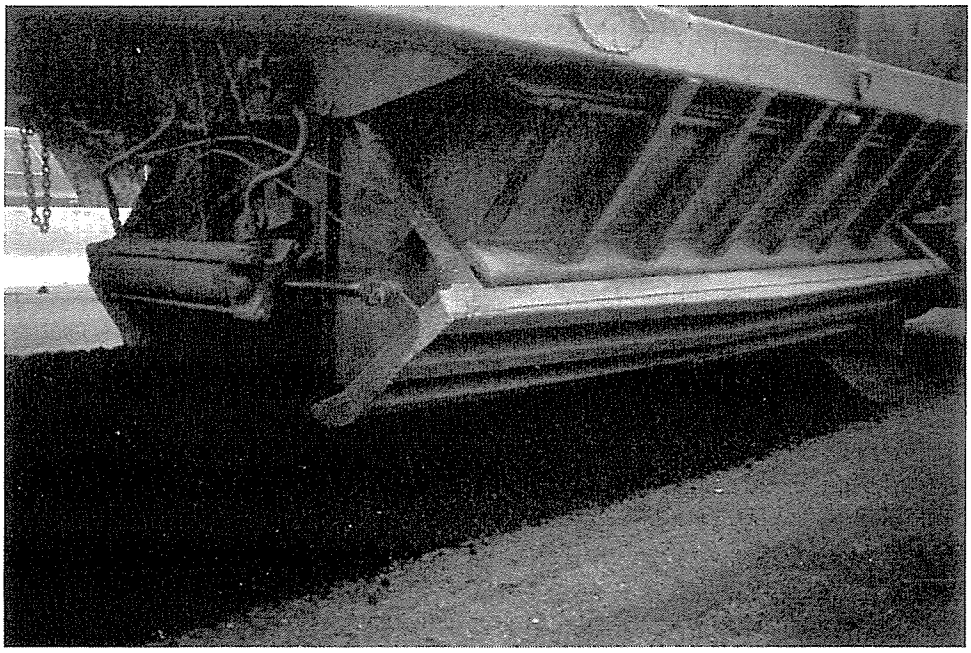


Figure 8.4 Bottom Dump Trailer

each time the operator finishes dumping a load onto the grade. Cleaning is important; the hauling unit should empty with no chunks of material stuck to the sides or the bottom of the hauling unit. The inside of the trailer should be coated with an approved release agent. The doors should be lubricated daily so that they react on the command of the dump operator.

Bottom- or belly-dump trucks generally require that a dump operator be stationed on the grade to open the gates and control the spread rate of the HMA material onto the grade in front of the paver. The dump operator is responsible for maintaining the proper amount of material to keep the paver hopper uniformly charged. During paving, if the quantity on the grade is found to not match the paving width or depth, material should be added or removed—without stopping the paver.

Horizontal-Discharge Truck

Horizontal-discharge hauling units have a conveyor system that moves the material from the hauling unit into the paver hopper without the need to raise the truck bed (see Figure 8.5). This system has a safety advantage because the bed does not need to be raised, and segregation is minimized as the material is moved in mass on the conveyor system as it is discharged into the paver hopper or material transfer device (described later). Appropriate maintenance procedures must be followed for these systems to operate properly. Proper cleaning of the conveyor ensures that cold chunks are not left in the truck bed. Daily checks between loads ensure that the conveyor system is operational and that no loose or damaged components will break during the workday.



Figure 8.5 Horizontal Discharge Truck

Avoiding Heat Loss

The temperature of the HMA must be maintained within specified limits during production, storage, transportation, and laydown. Loaded hauling units must deliver the HMA with minimal heat loss. If delivery time to the paving site is such that the material cools below acceptable placement and compaction temperatures—or if adverse conditions are such that the material will cool in the hauling unit—steps must be taken to retain the heat. Tarps and insulated truck beds are commonly used to reduce heat loss. The temperature of the HMA in the hauling unit should be monitored. If the temperature fluctuates beyond the limits allowed in the specifications, steps must be taken to reduce or eliminate the excessive temperature variations.

Condition of Haul Trucks

Numerous styles of haul trucks exist today, each with unique maintenance procedures. Some basic maintenance procedures, however, apply to all styles. The dumping lip should be cleaned with a scraper between loads, and tailgate-latching devices should be lubricated properly so they release on command. An approved release agent must be applied to the inside of the hauling unit before the next load is dumped into the hauling unit. The floor of the hauling unit should be flat and free of dimples that might cause the release agent to puddle. The truck should be in good condition and capable of safely and legally transporting material to the job site. The lifting device should be maintained so that the bed can be lifted to continuously move the material from the truck bed to the paver hopper.

All hauling units should be continually monitored for signs of mechanical problems or leaking fluids (except air conditioning condensate). If problems are found, the hauling unit should be immediately removed from the lineup until permanent repairs can be made.

8.3.2 Delivery of HMA to the Paver

When the loaded hauling unit has arrived at the paving site, the HMA must be moved into the paver without disturbing the HMA gradation or without losing temperature. The driver, after arriving in the staging area, should "break the load" by raising the truck bed slightly until the HMA moves against the tailgate. This procedure allows the HMA to exit the truck bed in a mass rather than in a dribble, avoiding segregation as the HMA flows into the hopper (see Figure 8.6).

The paver operator or dump operator, should position the loaded truck 12–18 inches (300–450 millimeters) short of the paver, with the dump bed partially raised. This positioning keeps the truck from knocking the paver backward and creating an indentation in the mat. As the paver moves forward and makes contact with the hauling unit, the driver should be instructed to release the tailgate and raise the dump bed as quickly as possible. When the tailgate opens, the material will move in mass into the hopper, thereby reducing the chances of segregating the material. When a wheel-locking device is not being used, the experienced truck driver will anticipate this sudden rush of material and briefly apply additional brake pressure to prevent the hauling unit from being forced out of the hopper and spilling material on the pavement directly ahead of the paver. While the dump bed is emptying, the driver should maintain just enough pressure on the brakes to keep the truck from rolling away from the paver. The flow of HMA is coordinated between the paver and truck driver. The bed should be raised to the fully extended position as quickly as possible without overflowing the paver hopper. Once the bed is empty, the paver signals for the truck bed to be lowered before pulling out of the paver hopper. The hauling unit should be driven to a predetermined cleanout area, where remaining HMA can be removed from the truck bed and around the tailgate area and apron. Do not dump HMA in front of the paver or allow beds and aprons to be cleaned in front of the paver. See Figure 8.7 for the steps associated with properly loading the paver.

8.3.3 Control of Material Surge in Front of Paver

Material Transfer Device

Experience has shown that continuous paving that minimizes stopping with a properly set-up and operating paver will produce the smoothest



Figure 8.6 Proper Method for Breaking the Load

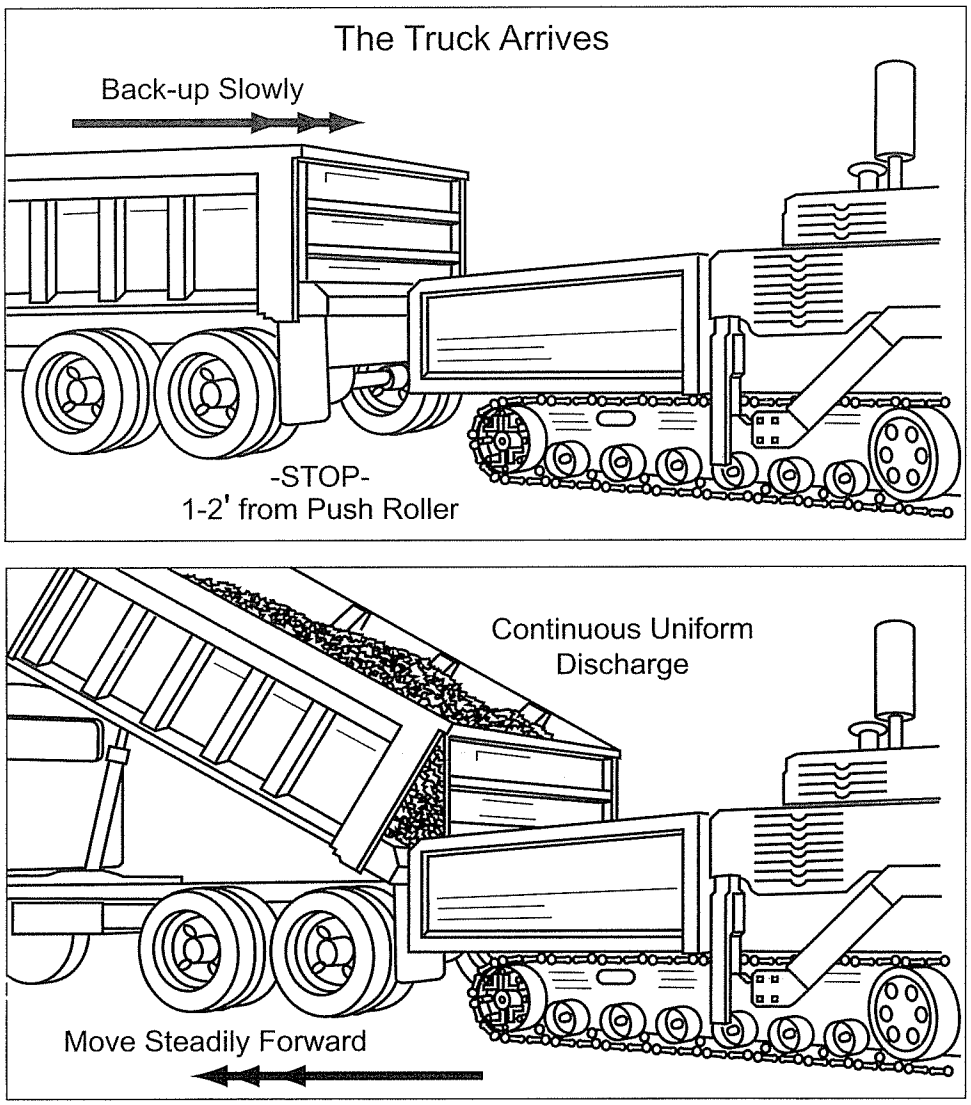


Figure 8.7 Proper Loading of Paver

HMA pavement possible. A material transfer device (MTD) makes a non-stop paving operation possible. The material holding bin on an MTD acts as a surge hopper and can hold up to 25 tons (23 tonnes). A special insert in the paver hopper holds an additional 15 tons (14 tonnes). With the equivalent of two to three truckloads of material immediately in front of the paver, this "surge of material" takes up the slack between deliveries and keeps the paver moving forward until the next hauling unit arrives (see Figure 8.8).

MTD Remixing Ability

Today's MTD can reblend the asphalt mixture to minimize both mixture and temperature segregation as the mix passes through the machine. Material is re-blended at different points within a machine depending on



Figure 8.8 Material Transfer Device

the manufacturer. One manufacturer uses a triple-pitched auger located at the bottom of the MTD's surge bin. The auger moves the material from the outside edges toward the center, where the material is dropped onto a conveyor that moves it to the discharge conveyor. Another manufacturer has placed a small reblending pugmill at the discharge opening of the hopper insert. The material is reblended just before it drops onto the paver's material conveyor system. Another manufacturer uses mixing augers in place of slat conveyors to move and reblend the material from the hopper to the discharge conveyor of the MTD.

Windrow Paving

Windrow paving places the HMA in a longitudinal windrow directly on the grade in front of the paver as it arrives on the project, and as with the MTD, this surge aids in achieving a nonstop paving operation. An elevating windrow device or pickup machine (see Figure 8.9) in front of the paver picks up the material and places it in the hopper. The quantity of HMA on the grade is controlled by a dump operator as the hauling unit is unloaded and the size of the windrow is adjusted according to the level of material in the hopper. A well trained crew can successfully pave for hours, producing a high quality HMA pavement without stopping.

Windrow Paving—Remixing Ability

The windrow-building process can allow the mixture to segregate with the larger aggregate accumulating along the bottom outside edges of the windrow. Therefore, steps must be taken to ensure that the windrow pickup device picks up the entire windrow cleanly and that the segregated



Figure 8.9 Windrow Paving

material is directed back toward the center of the material flow as it is moved through the windrow pickup device. Adding plates to force the material into a smaller stream and keeping the volume of material in the hopper at a high level help reduce mix segregation.

Advantages and Disadvantages of Windrow Paving

The advantage to windrow paving is the continuous aspect of the operation, allowing the paver to lay a consistent high quality HMA pavement. The problems can be heat loss into the grade, the inability to size the windrow, the inability to cleanly pick up the material off the grade, and finally vibrations caused by the powered pickup machine that affect the paver's performance. All these issues can be overcome with proper attention to detail. A shorter windrow will reduce heat loss; proper laydown crew training about windrow sizing will solve quantity problems; and a properly adjusted pickup machine, isolated from the paver, will pick up the material cleanly without adversely affecting the paver. Windrow paving operations are usually confined to warm dry climates, where the chance of a surprise rainstorm cooling the material in the exposed windrow is minimal.

8.3.4 Load Tickets

Load tickets are a written record of deliveries to the job site. They detail the type, quantity, and temperature of the delivered HMA, the time of load

out, and the identity of the truck and the driver. Tickets indicating the amount of hot mix asphalt placed on each driver's truck and the accumulated total shipped to the current paving operation to that point are given to the truck driver before the truck leaves the HMA mixing plant. Collection of the tickets at the project ensures that the material represented by the load ticket was actually incorporated into the work. An accurate record of all load tickets minimizes the possibility of any loads being diverted from the job. They are also used as a check against plant records for the day's total tonnage. With the collection of load tickets and the desired spread in pounds per linear foot (kilograms per linear meter) or tons (tonnes) per station, the average spread rate per load can be checked. The Material Safety Data Sheet information required by OSHA can be preprinted on the back of the delivery ticket.

8.3.5 Visual Inspection of the HMA

Inspection of the HMA as it flows from the hauling units into the MTD, into the windrow in front of the paver, or directly into the paver hopper will help determine the consistency (temperature, gradation, aggregate coating, asphalt content) of the HMA. Observing how the HMA is laying in the hauling unit reveals load inconsistencies. Watching the material as it breaks out of the hauling unit is also important—one load may break uniformly as the bed is raised while the next load might break in larger chunks or move all at once into the hopper, windrow, or MTD. Uncoated aggregate is easily identified and shows that the material has not been properly mixed during production. A visible fume as the HMA is moving out of the hauling unit is a normal condition. But if the level of fumes is inconsistent from one load to the next, the HMA temperature might be fluctuating and should be checked for compliance. The temperature of the HMA should be kept to $\pm 25°F$ ($\pm 15°C$). Finally, segregation should be noted, especially at the beginning and at the end of each load. If end-of-load segregation is apparent, the loadout techniques at the HMA plant should be adjusted to reduce or eliminate end-of-load segregation.

8.4 Paving Equipment

HMA pavers in use today differ in size and detail with many intricate parts and adjustments, but all have similar operating principles. The paver receives the HMA directly from the hauling unit, MTD, or windrow pickup machine and moves the HMA through the paver to place and spread the HMA on the roadway before beginning the compaction process (see Figure 8.10). Figure 8.11 traces the flow of asphalt mix from the receiving hopper at the front of the paver to the finished pavement behind the screed at the rear of the machine. The paver spreads the mixture in a uniform layer of desired thickness and shape, or finishes the layer to the desired elevation and cross section, making it ready for compaction. Modern pavers are supported on crawler treads or pneumatic-tire wheels. These machines can place a layer of less than 1 inch (25 millimeters) to approximately 10 inches (250 millimeters) in thickness over a width of 6 to 32 feet (2 to 10 meters) Working speeds generally range from 10 to 70 feet (3 to 20 meters) per minute.

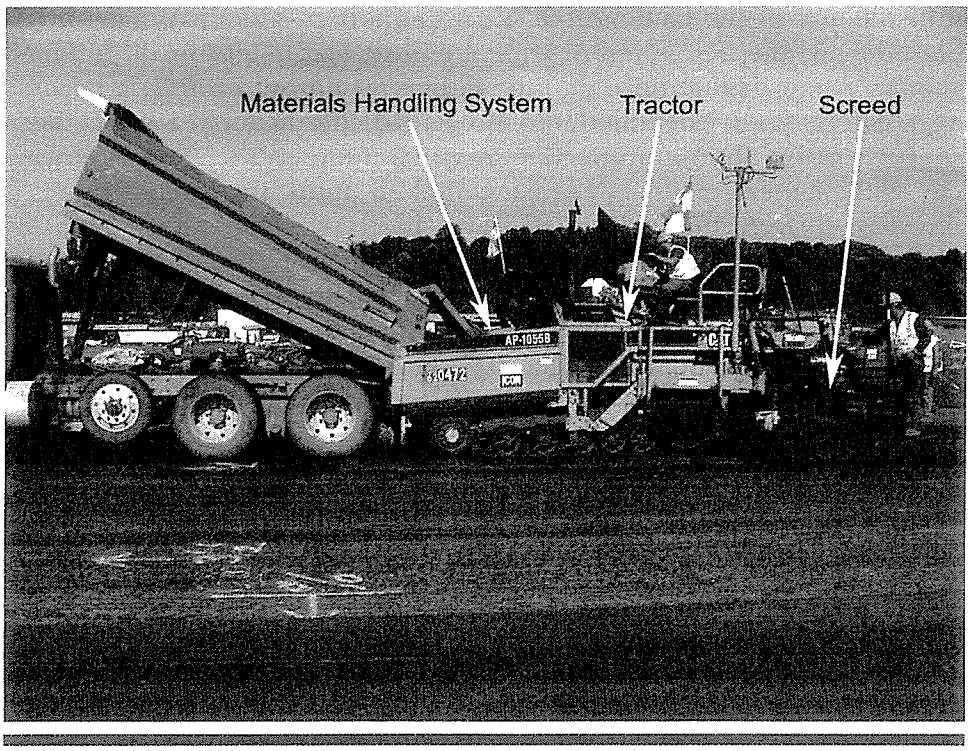


Figure 8.10 Asphalt Paving Machine

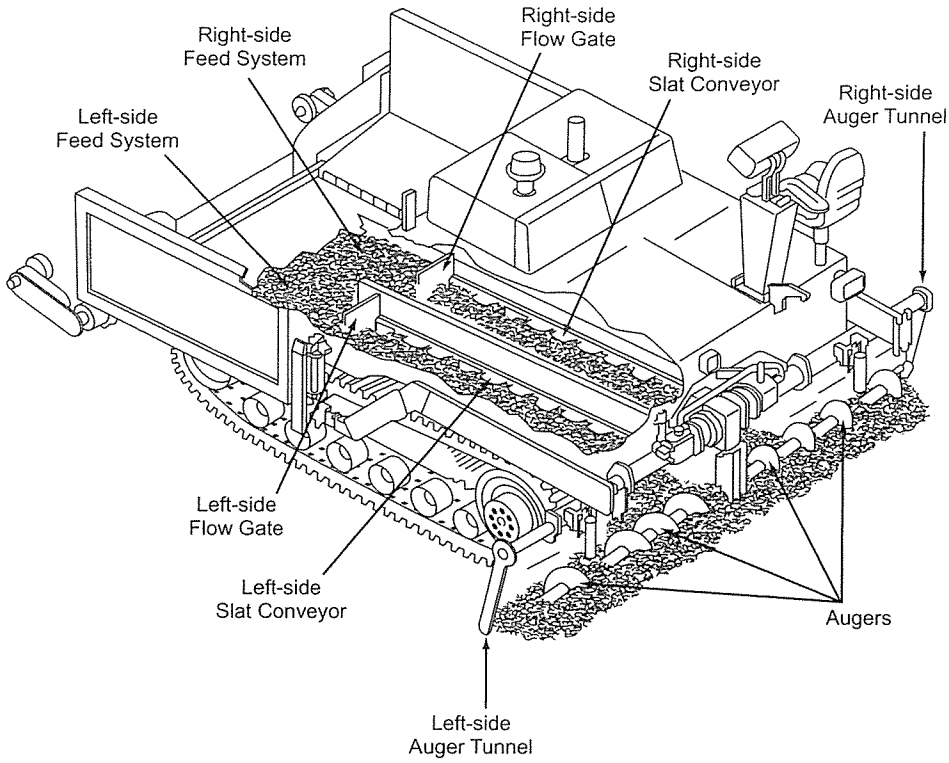


Figure 8.11 Schematic of Asphalt Paver

8.4.1 Asphalt Paving Machine—Basics

An HMA paver is a self contained, self-propelled pavement finishing machine made up of three basic units: the tractor, a material handling system, and the screed. The tractor portion of the asphalt paver contains the power unit that provides the power needs of all three components, plus the drive train that propels the machine. The material handling system receives the HMA material and moves it from the hopper to the spreading augers in front of the screed. The screed levels and profiles the material being laid, provides some initial density, and seals the surface ready to be compacted to the final specified density as described in Chapter 9, "HMA Compaction."

Identification of Components of the Paver

The following is a description of each component of the asphalt paver:

- **Hopper.** The hopper is the area where the HMA is deposited by the hauling unit, pickup machines, or material transfer device.
- **Hopper insert.** The hopper insert is a metal bin inserted inside the paver hopper to receive material from the MTD or a windrow pickup machine to expand the hopper storage capacity by 50 to 70 percent (see Figure 8.12).
- **Push rollers.** These rollers mounted on the front of the paver push on the hauling unit tires as the HMA is being transferred from the hauling unit to the paver hopper.



Figure 8.12 Rubber-Tire Laydown Machine with a Hopper Insert

- **Slat conveyer.** The slat conveyer is the mechanism used to move the HMA from the hopper to the augers in front of the screed. The conveyor gets its name from the thin flat metal bars connected at both ends to two parallel chains that pull the mix from the hopper. One manufacturer has replaced the slat conveyor with two longitudinally mounted screw conveyors.
- **Flow Control Gates.** These adjustable metal plates change the size of the opening at the hopper discharge used to control the amount of material flowing out of the hopper.
- **Tow point.** This point near the center of the paver where the side arms are attached is a pivoting connection that allows the side arms to rotate around this point.
- **Side arms.** The side arms connect the screed to the tow point on the tractor and are responsible for towing the screed.
- **Auger.** The augers receive the material from the slat conveyers, moving it transversely and uniformly across the full width of the screed.
- **Screed.** The screed is the most important component of the asphalt paver. It is essentially a wide flat piece of heated steel that knocks down and smoothes out the mass of material placed in front of it by the augers. In doing so, it places the material to the proper grade, cross slope, and texture and provides initial compaction of the HMA. The use of a screed extension allows paving widths wider than the basic screed provides.
- **Depth-Screws.** These manually operated screw mechanisms change the angle of the screed plate as it passes through the HMA

material, allowing more or less material to pass under the screed. The careful control of this angle, also known as "angle of attack," precisely controls the thickness of the mat being placed.

8.4.2 Types of Tractors

Only two types of drive mechanisms power today's asphalt pavers—either a track or a rubber-tire propulsion system (see Figures 8.12 and 8.13).

The selection of either a rubber-tire or a track-mounted paver is up to the individual user. Either unit will provide the uniform forward speed required to lay smooth asphalt pavements. A track system provides a lower ground pressure with greater traction, while a rubber-tire paver has greater mobility and a lower operating cost. The mobility of track machines has been improved by the introduction of a continuous rubber-track system. On rubber-tire pavers, both drive tires should be inspected each morning for cuts, punctures, and other damage that might result in a flat tire during the work shift. To achieve maximum traction from drive tires, most are filled with calcium chloride or some other weight-producing material. The air pressure must be closely monitored and checked with special pressure gauges and should never vary by more than 2 pounds per square inch (13.8 kilopascals) between the two drive tires. Caution must be exercised when checking the air pressure in liquid-filled tires to avoid the liquid's spraying out and coming in contact with the skin and eyes. Track systems on track pavers should also be checked daily to ensure that they properly align, have correct tension, and have no gouges in the track pads.



Figure 8.13 Track-Mounted Laydown Machine

Material Handling System

The material handling system consists of the mechanisms used to move the material from the hopper to the augers in front of the screed. Slat conveyers are used on most pavers to move the material through the paver. One paver manufacturer uses four counterrotating augers in place of slat conveyers to remix the HMA to minimize segregation as the material moves from the hopper to the augers. The volume of HMA required to match the width, depth, and speed of placement is precisely controlled either by flow gates mounted at the rear of the hopper or by varying the speed of the slat conveyer system. When flow gates are not used, the drives of both the auger and the conveyer are separated and controlled independently to transfer the correct volume of HMA. Split systems use an additional material sensor to control the volume of material carried by the conveyer into the auger area. As the HMA flows into the area in front of the screed, the augers move it transversely to the outer ends of the screed. The material sensors at the outer ends of the screed control the amount of HMA in this area by speeding up, slowing down, or stopping the rotation of the augers. The auger systems on most pavers can be raised or lowered to provide clearance for loading and unloading the paver from equipment hauling trailers. This vertical adjustment can also be used when paving deeper lifts of HMA. The augers should always be 1½ or 2 inches (38 to 50 millimeters) above the HMA pavement thickness being placed. When paving is done with a screed wider than the basic screed, auger extensions and material confining plates known as tunnel extensions should be added to uniformly carry the HMA the full width of the screed. Auger tunnel extensions (slide-out or bolt-on) are located in front of the augers when running at extended widths (14 feet [4 meters] or more). Ideally, the auger tunnel extensions should be 12–18 inches (0.3–0.5 meters) from the edger plate of the screed to minimize the chance that the material will segregate.

The volume of material in front of the screed is automatically controlled by material-control sensors that signal the material-feed system to deliver the proper amount of HMA to the front of the screed. These sensors should be properly positioned to make sure that the volume is not allowed to fluctuate beyond operational limits in achieving a quality mat.

Utilization of the Material-Control Sensors

The material-control sensors precisely control the volume of material (head of material) in front of the screed. The augers on the right and on the left are independent of each other and are controlled by separate material-control sensors. The sensors control the movement of material by starting and stopping and running at variable speeds in order to furnish the correct volume of material to match the paver's width, depth, and speed.

To operate properly, the sensors must index from the live head of material that is moving constantly to fill in an empty area in front of the screed. Paver manufacturers use different types of sensors and mount them in different positions to monitor and control the flow of material. Two types of material sensors are used today: contact sensors and ultrasonic sensors. Contact

sensors use a switching device, such as a potentiometer or a three-stage switch attached to a paddle physically touching the flow of material, which signals the auger drives to start, stop, or vary the speed of the augers (see Figure 8.14). Ultrasonic sensors are used the same way, except they are a noncontact devices that use sonic sound waves to measure the volume of material in front of the screed (see Figure 8.15).

When a balanced screed (equal distances from the center of the paver to the outside edges of the screed) is being run, the paver is much easier to set up and operate. If the screed is wider or paving deeper on one side, a greater amount of HMA is required on that side. In this unbalanced flow condition, fine tuning the flow-control gates or adjusting the speed of the individual material conveyors is critical.

8.4.3 Screeds

Screed Components and Adjustments

Crown

The main screed has the ability to be crowned at the center of the screed. With this crown, the screed can be set up to give either a positive or a negative crown to the profile of the HMA pavement being laid. Positive crown is used when the centerline of the screed is intended to be the centerline of the road and when fall or drainage is required in both directions. Negative crown is used if the center of the HMA mat being laid is used for drainage.

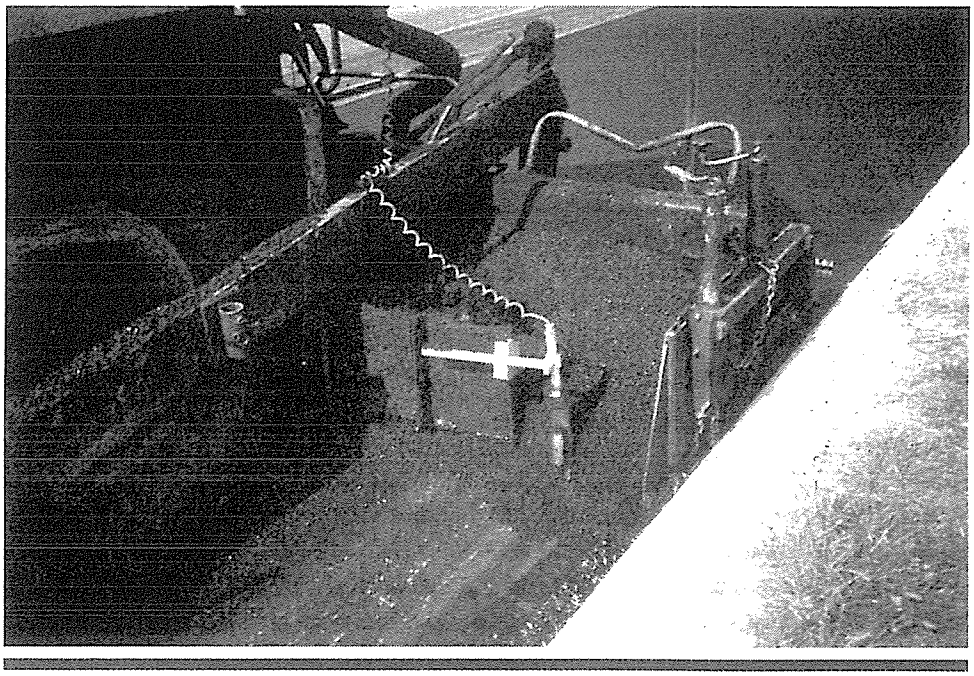


Figure 8.14 Contact Sensor for Material Control

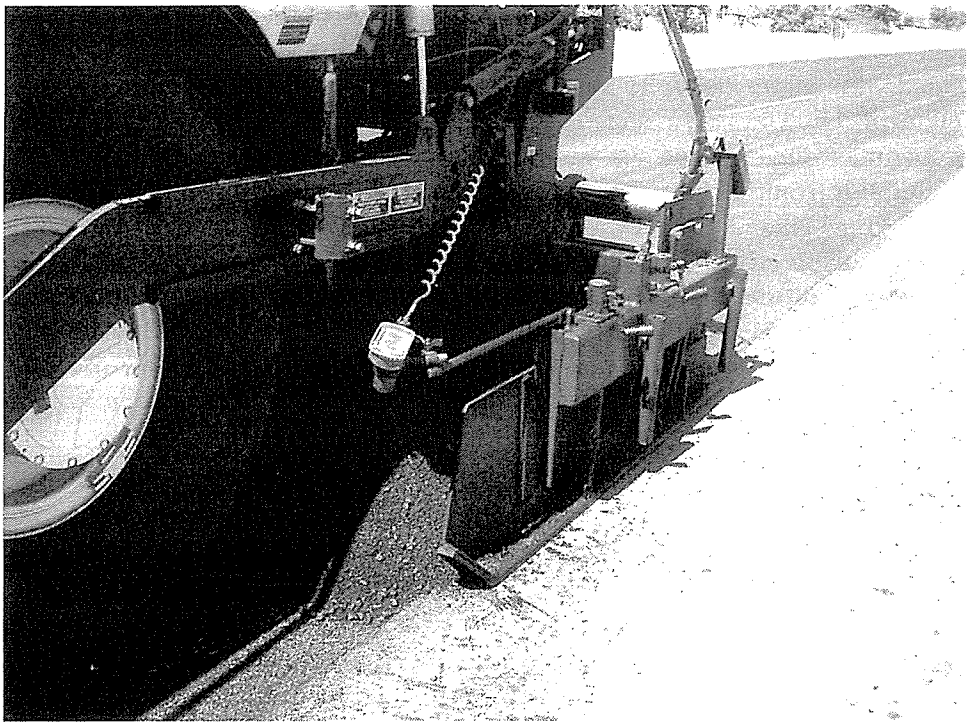


Figure 8.15 Ultrasonic Noncontact Sensor for Material Control

Vibratory and Tamper Bar Systems

Vibrators on the asphalt screed provide some initial density. The compactive force, which is generated by offset weights as they are spun at high rates around a central shaft, is applied to the HMA as it passes under the screed. Most vibrators on today's pavers are adjustable. The vibrators should not be used when paving a thin lift because fracturing of the aggregates might result. A nuclear density gauge can be used to check behind the screed at various vibrator settings to evaluate the density. If density does not increase, the vibrators should be turned off. This trial-and-error process helps establish the correct setting for each combination of HMA mixture and lift thickness and should be done on each new mixture that is required to meet a density specification.

Some screeds have tamper bars at the leading edge of the screed that serve a dual purpose. They strike off the HMA as it passes under the screed and give a greater initial density than a vibratory screed. The only shortcoming to the tamper-bar system is that higher rates of paving speed do not give the tamper bars enough time to tuck the material under the screed. This can result in a torn HMA pavement surface or one with a dragging texture. The tamper bar should protrude below the leading edge of the screed just enough to catch a fingernail on it, or $\frac{1}{64}$–$\frac{1}{32}$ inch (0.4 to 0.8 millimeter). Dual tamper-bar screeds have one tamper bar in back of the other at the leading edge of the screed. This second tamper bar gives

additional initial density to the HMA pavement being laid. This system must be vigorously maintained to operate properly. A release agent should be applied when the machine is shut down at the end of a shift. Failure to maintain the tamper area as suggested by the manufacturer results in accelerated wear and a poor quality HMA pavement texture.

Screed Heating Systems

At the start of a paving shift, the screed is much colder than the HMA being laid. To keep the asphalt material from sticking to the colder metal of the screed, it must be preheated. Two types of heating systems are used to preheat the screed before each day's paving: diesel and electrical. Diesel burners are simple and easy to use, but they require proper operational procedures so that excess fumes are not produced when air and fuel in the burner chamber are out of balance. Electrical heating systems have heating elements immediately above the screed plate that are surrounded by special heat transfer oil that helps heat the screed uniformly without hotspots. This system minimizes fumes and provides a more even heat across the entire screed plate. Electrical heating systems operate in a couple of different modes. In one, the screed is heated to operational temperature and the heating system shuts off. The other heats the screed to the desired temperature and cycles on and off to maintain this temperature throughout the paving operation.

Free-Floating Screed

Free-floating screeds (see Figure 8.16) have been a standard feature of HMA pavers since they were first introduced in the 1930s. The basic principle of the free-floating screed is that, when it is pulled into the material deposited in front of it by the augers, it will automatically ride up or down seeking the level where the path of its flat screed plate is parallel to the direction of pull. At this point, the forces (listed below), which act upon the screed, will be in balance and the screed will continue to lay a uniform mat thickness until one or more of the forces change.

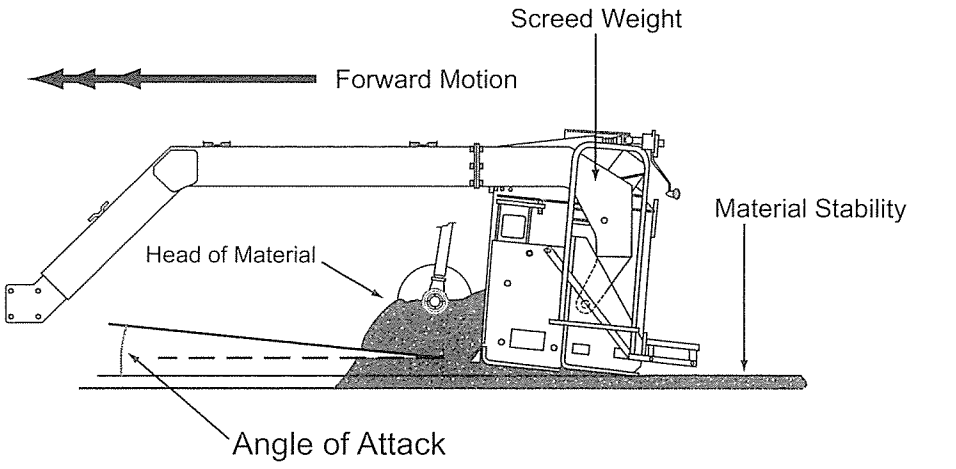


Figure 8.16 Free-Floating Screed

When a free-floating screed is used, five factors must be considered:

- The angle of attack
- The head of material
- The forward motion (paving speed)
- The screed weight
- HMA stability (temperature and gradation)

Angle of Attack

The angle of attack is the angle of the screed in relation to the paving grade. The screed pivots about the screed pivot point. The angle can be adjusted by the depth-screw assembly or by raising or lowering the screed tow point. The depth-screw is adjusted manually by the screed operator. Tow-point changes are generated by automatic grade and slope control. The angle of attack creates an extrusion angle under the screed; once the extrusion angle can support its weight, the screed is free to float at that average depth. Pavers have a natural leveling effect, placing asphalt thinner over the highs and filling in the lows of the base. Increasing the angle of attack allows more material to flow under the screed, causing the screed to rise until it reaches the elevation that finds equilibrium of the forces that act upon the screed. The paver will continue to place HMA at the newly established depth until one or more of the forces acting on the screed is changed. Reducing the angle of attack results in less material passing under the screed, and the screed will drop until the new depth is reached and a new equilibrium of forces is established.

Head of Material

The head of material is the depth of material being evenly spread in front of the screed by the auger. Controlling the head of material is the most important factor in laying a smooth HMA pavement. The head of material is affected by the volume of the HMA in the auger chamber. Once the head of material has been established, it must be maintained within a tolerance of ± 1 inch at all times. Starving the screed of material will result in the screed settling and will cause a dip in the pavement. Immediately placing a large amount of HMA in front of the screed will guarantee a bump as the screed attempts to crawl over the excess material deposited in its path. Ideally, the head of material should be maintained at a height that exposes one half of the auger and not vary by more than ± 1 inch (25 millimeters).

Paving Speed

Nonstop paving at a uniform forward speed results in the highest-quality and smoothest pavements and should be the goal of every paving crew. Changing the paving speed will change the angle of attack on the screed, causing a corresponding change in the mat thickness. Manual adjustments to offset this depth change must be made with the depth screws. When equipped with automatic controls, these devices immediately sense the change in depth caused by speed changes and make the appropriate adjustments to maintain the required depth. Normal operations require that

the paver be stopped occasionally. When the paver must be stopped, a rapid stop-start method should be used. This means that the paver should be stopped and started as quickly as possible without a gradual slowdown or speed-up period, which limits the fall or rise of the screed.

Screed Weight

The screed weight is usually translated into pounds per square inch (kilograms per square meter), or the vertical force applied by the screed into the HMA mixture being laid. This unit weight is constant except when extending or retracting the variable-width screed. Care must be taken to keep the head of material from fluctuating when extending the screed. When the screed is operated manually, a slight change in the angle of attack may need to be made by using the screed-depth controls to maintain the desired HMA pavement thickness. If automatic grade and slope controls are being used, the angle of attack will automatically change. Using the manual override on the auger controls can help minimize the screed-depth changes. When extending the screed width, the augers should be controlled manually to supply a constant volume of material in front of the screed extension. When the extension is retracted, the auger should be manually shut off to reduce the amount of material in front of the extension so that it can be retracted without causing the screed to rise.

HMA Stiffness (Loose)

A change in the temperature of the HMA or the gradation of the aggregates can result in a change in the stability or stiffness of the loose HMA during paving. This change in stiffness will cause the screed to either rise or fall as it reacts to a change in vertical force and seeks to rebalance all of the forces acting upon it.

Screed Extensions

Screed extensions are rigidly mounted, fixed-width, or hydraulically powered variable-width attachments that extend beyond the width of the main screed to facilitate paving widths that are wider than the basic machine width. Both types of extensions provide full screed plates with heat and compaction similar to the cross section of the main screed. This allows the HMA placed by the extension to have the same characteristics in terms of texture, smoothness, and precompaction as the material placed by the main screed. Both types of extensions can be angled differently than the main screed, allowing adjacent lanes and shoulders to be paved at a different cross-slope than the main screed.

Variable-Width Screeds

With a variable-width screed, the paving width can be changed with the push of a button while still maintaining a uniform surface across the entire HMA mat being placed (see Figure 8.17). Extensions on the variable-width screeds can be either front- or rear-mounted and will require different adjustments to achieve a quality HMA pavement. The extensions can usually be sloped positively or negatively at the outer edges of the main screed so the adjacent lane(s) can be paved at the same time as the mainline.

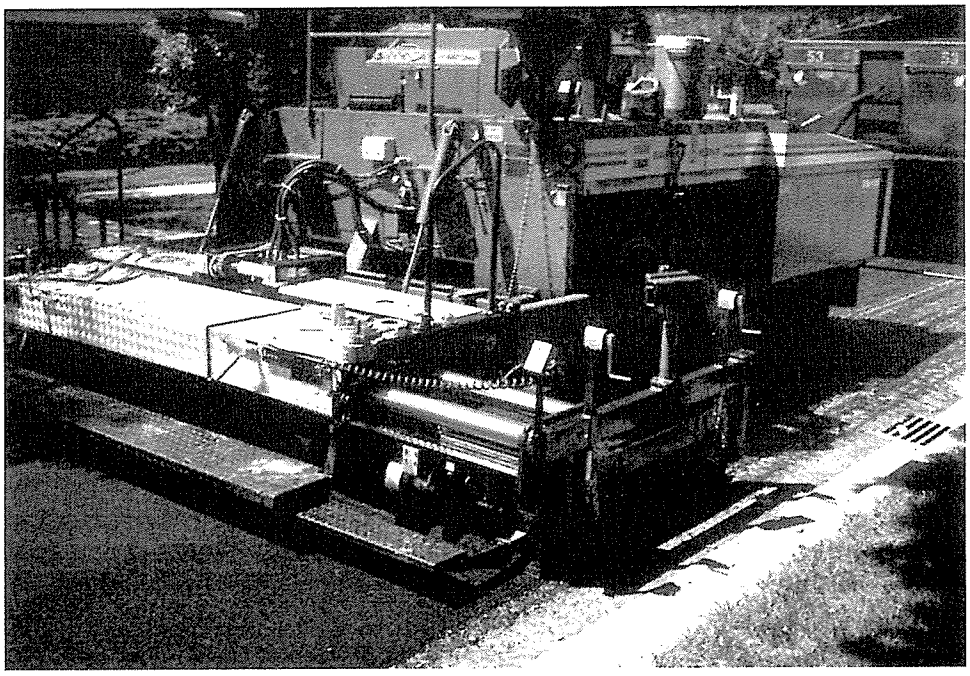


Figure 8.17 Screed with Variable-Width Extensions

It is not unusual to pave a 4- to 10-foot (1.3–3.3 meter) shoulder of a major highway while laying the 12-foot (3.7 meter) mainline.

Screeds with Fixed-Width, Bolt-on Extensions

Fixed-width screeds with rigidly fixed extensions are used for extended-width paving when a fixed width is to be paved without any variations in the required width (see Figure 8.18). They have fewer moving parts and thus require minimal adjustments in order to perform within expectations. Fixed screeds begin with a basic 8- or 10-foot (2.4- or 3.0-meter) screed, and rigidly fixed extensions of various widths are added to extend one or both sides of the screed to the desired paving width. The structural stability of the fixed-width screed offers the ability to pave much wider widths and maintain better control of the desired profile. Screeds with rigidly fixed extensions cannot change paving width unless the paving operation is halted long enough to add or detach the extension(s) needed to change the required paving width. Rigidly fixed extended screeds are best suited to long stretches of fixed paving widths.

Hydraulic Gates

Commercial paving, such as in parking lots, often requires working in and around obstructions. The ability to extend and retract a screed extension reduces the handwork needed to complete a project. The use of a simple hydraulic gate arrangement makes this possible without the added weight and complexity of a full screed extension. The hydraulic gate is a simple extension with a short strike-off plate attached to the basic screed powered in and out with a hydraulic cylinder. Because there is no heat

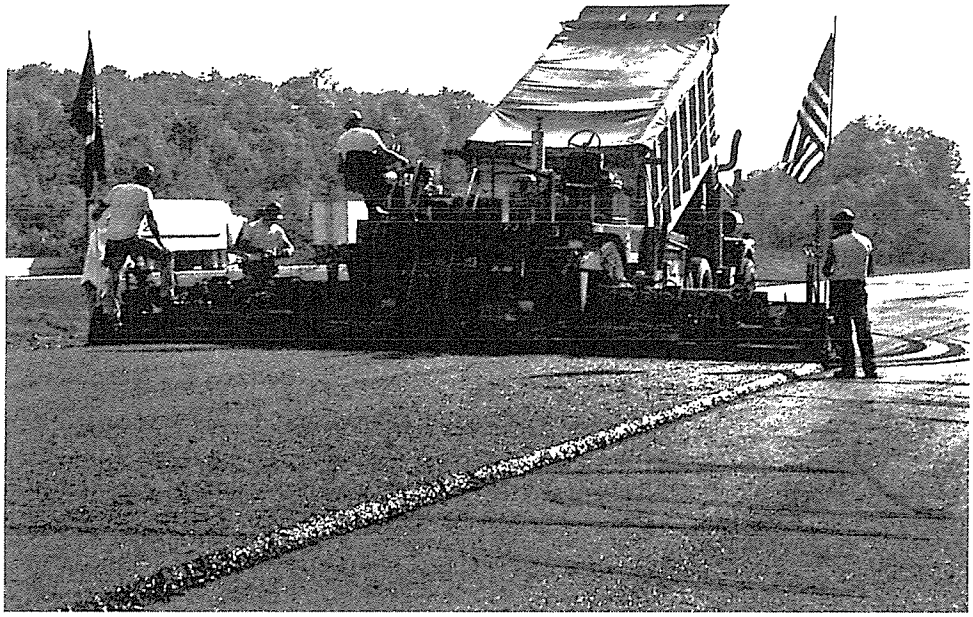


Figure 8.18 Screed with Fixed-Width Extensions

or pre-compaction vibration on the extension, it must be run above the surface left by the basic screed. This will compensate for the lack of initial compaction achieved by the basic screed. Once the HMA pavement has been compacted, the two surfaces will match. Some hydraulic gates have a wider strike-off assembly, which can be set to allow some extrusion by means of an angle of attack established in the strike-off. This style of gate must be set so the surface laid by the gate matches the one laid by the basic screed. Because of their design and operation, the use of hydraulic gate screed extensions should be limited to commercial or shoulder paving and not allowed on mainline paving.

Adjusting Extensions on Variable-Width Screeds

Front- and rear-extendable screeds are widely used. These screeds have numerous adjustments incorporated into them to produce high-quality asphalt pavements. Some adjustments are very specific to the design of the screed, and the manufacturer's manual should be checked for procedures for adjustment and maintenance. The objective in operating an extendable screed with either front- or rear-mounted extensions is to maintain the screed plate surface in full contact with the HMA pavement being laid at all times. This will produce an HMA pavement with a consistent texture from side to side that is also free of blemishes and steps where the main screed and extension meet. To do this, the extensions must be perfectly aligned with the main screed. This adjustment is the same for both front- and rear-mounted extensions. The main screed should be placed on two

6-inch (150 millimeters) blocks about 3 feet (1 meter) long at each quarter point. The extensions should be retracted fully and using a level or other straight edge, checked that the extensions are flat with the main screed. If not, adjustments should be made until the main screed and extension are flat with each other. The screed should be fully extended while the operator watches closely to ensure that the extension remains aligned during the full extension. The extension should remain flat and aligned with the main screed from zero width to full extension. This process should be completed on both sides.

If a step-up or misalignment (see Figure 8.19) is observed at the junction of the main screed and extension, the extension must be raised or lowered until the discrepancies disappear. If the HMA pavement being laid by the extension has a shiny texture compared with the rest of the mat, the angle of attack on the extension may have to be slightly reduced (according to the manufacturer's recommendations). Another way of checking HMA pavement texture differential is with a density gauge. If the density behind the screed extension is slightly different than the density behind the main screed, the angle of attack on the extension should be adjusted and the density rechecked until all surfaces behind the screed are of similar density and texture.

Front- and rear-mounted screeds vary in their ability to collapse the screed to its minimum paving width. Regardless of screed type, HMA material must be removed from in front of the extension in order to retract the screed. Failure to remove this material will prevent retracting the screed to minimum width.

8.4.4 Automatic Screed Controls (Types and Operational Differences)

Automatic Grade Controls

Automatic grade controls take a continuous reading from a known longitudinal reference and transfer it to the paver's control system, which continually changes the depth of the HMA pavement being placed. The controls are either mechanical or electronic.

The mechanical device uses a spring-loaded rotational potentiometer attached to a wand as a sensor to measure changes in the vertical distance between the reference line and the sensor mounted on the side arm of the paver. In response to these changes, the sensors send a signal proportional to the amount and direction of change to the paver's grade controls. The grade control then actuates the hydraulic cylinder, which makes a corresponding change in the height of the tow point. On a free-floating screed, as the tow point rises or falls, the angle of attack changes and the depth increases or decreases accordingly. The screed changes depth until the distance between the reference and the sensor is reestablished.

Electronic control equipment can also be used to automate the screed. Instead of using a mechanical device for measuring the change in vertical



Figure 8.19 Improperly Aligned Screed Extension

distance between the sensor and the established reference, electronic control equipments uses sound waves. The wave signal is emitted, and the time it takes the signal to hit the established reference and return to the sensor is converted into a distance. Electronic computers can also be used to automate the screed. The thickness of the overlay or the proposed elevations of the new pavement are computed along the length of the project or across a grid for paving large areas and then programmed into the screed-controlling computer. As the paver moves forward, it tracks its position with input from an electronic measuring wheel, a total station theodolite, or the satellite-driven Global Positioning System. Screed settings are adjusted automatically to achieve the desired thickness or elevation. Because this system attempts to achieve a final thickness or elevation, a rolldown or compaction factor (that tells the computer how much height will be lost going from the uncompacted to the fully compacted mixture) must be programmed into the computer.

Automatic Slope Control

An HMA paver equipped with an automatic slope control can hold the screed at a different cross-slope from the slope of the surface that the paver is riding on. The slope control uses a pendulum or bubble level device to establish a perfectly vertical reference, from which a perpendicular plane will establish a true horizontal line. Using this true horizontal line, a cross-slope can be created and maintained across the screed width to build pavements with a cross-slope, such as superelevated pavements on curved sections of roadway. Because the slope can be adjusted while the paver is moving, the cross-slope can be "dialed" in or out and held constant as the paver enters and exits a transition from flat to superelevated pavements. When automatic slope control is used in conjunction with an automatic grade control setup, the side of the screed closest to the reference line will be the pivot side and the cross-slope will be set positively or negatively from this side as required by the project specifications.

References Available for Grade and Slope Control

Longitudinal reference lines for the automatic grade and slope-control sensors to follow allow a pavement to be built to a specific vertical and horizontal alignment. Fixed reference lines that are surveyed and set in advance of the paving operation are the most accurate and used mainly on new construction. Mobile references are devices attached to and travel along with the paver as they reference off of the existing surface that the paver is traveling on. Consequently, their use is limited to projects that have prepared surfaces that are acceptable to use as a reference. Whichever reference system is used, it is important to remember that the paver's automatic controls will follow the reference line accurately and closely duplicate the lines and grades in the HMA pavement. Consequently, the finished HMA pavement will be no better than the reference in terms of line, grade, or smoothness.

- **Stringline (grade wire).** The stringline (see Figure 8.20) is the ultimate control that can be used to build a pavement surface to an exact elevation. The stringline is a wire or nylon cord supported at intervals of 25 feet (8 meters) on metal posts and rods. The line is first anchored at one end of its length and then stretched very tightly to eliminate



Figure 8.20 Dual Stringline Reference Set Up for Airfield Paving

sags, which would be duplicated in the final pavement. One of the advantages to a stringline is that it provides a visual representation of the vertical grade line to be built into the pavement. By holding one's eye near the level of the stringline and sighting along its length, bumps, dips/sags, or even survey mistakes that will adversely affect the smoothness of the pavement can be seen and corrected. The stringline's drawback is the survey cost, erection, and maintenance of the grade reference throughout the construction period. The stringline should only be used if the result merits the expenditure.

- **Mobile references.** There are two types of mobile references: the rigid tube or pipe and the floating beam reference (see Figure 8.21). The purpose of the mobile reference is to average the effects of the grade changes in an existing reference surface over a greater distance than that obtained from just the wheelbase of the paver.
  - o The rigid tubular grade reference uses a rigid pipe that is 20, 30, or 40 feet (6, 9, or 12 meters) long. The tube rides on the existing pavement surface. A spring-loaded wire is stretched along the top of the pipe. The grade sensor on the paver that sends the electronic signal to the paver's tow point rides on top of the wire. As the pipe moves up and down over the existing grade, the stretched wire on the skid will average the differences in the elevation that occur under the mobile reference.
  - o A floating-beam mobile reference has a series of feet attached to the bottom of a beam. One or more of the feet can move over a single high or low point on the pavement surface without altering the slope of the